

Results Section Bundle:

Biological Risk Analysis for Lattice Radiotherapy

1 Results

1.1 Physical cohort

All 10 site-specific synthetic head-and-neck cases completed the physical transport workflow. Peripheral target coverage was stable across the cohort, with mean PTV D_{95} fixed at 3.50 Gy by design, while lattice selectivity varied more broadly. The median physical PVDR was approximately 1.80, and the physical cohort retained measurable spread in peri-tumour spill and OAR-adjacent dose. These distributions established the physical baseline against which biological reinterpretation was tested (Figure 1).

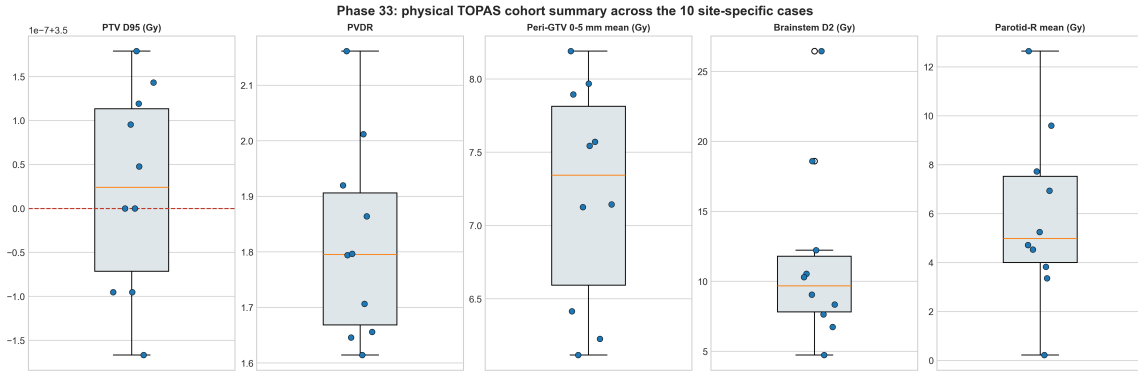


Figure 1: **Physical cohort summary.** Across 10 synthetic head-and-neck cases, the TOPAS-derived photon lattice plans achieved stable peripheral target coverage but retained broad variation in peak-valley contrast, spill, and OAR-adjacent physical dose.

1.2 Biological reinterpretation of the 10-case cohort

Applying the biology-informed post-processing substantially changed cohort interpretation. Relative to the physical-only view, cohort-mean PTV D_{95} increased from 3.50 Gy to 17.14 Gy in the no-sink analysis and 15.73 Gy with the anatomical sink included. In contrast, mean PVDR fell from 1.817 physically to 1.478 without sink uptake and 1.436 with sink uptake, indicating biological filling-in of the physical peak-valley pattern. Peri-GTV 0–5 mm mean increased from 7.219 Gy physically to 20.616 Gy without sink uptake and 19.929 Gy with sink uptake. Brainstem D_2 increased from 11.454 Gy physically to 28.399 Gy without sink uptake and 27.951 Gy with sink

uptake. Right parotid mean increased from 5.875 Gy physically to 18.821 Gy without sink uptake and 17.939 Gy with sink uptake (Table 1).

Table 1: **Cohort-mean endpoint summary across comparison modes.**

Endpoint	Physical-only	No-sink	Anatomical sink
PTV D_{95} (Gy)	3.500	17.144	15.731
PVDR	1.817	1.478	1.436
Peri-GTV 0–5 mm mean (Gy)	7.219	20.616	19.929
Peri-GTV 5–15 mm mean (Gy)	5.923	16.900	16.492
Brainstem D_2 (Gy)	11.454	28.399	27.951
Right parotid mean (Gy)	5.875	18.821	17.939

Biological reinterpretation changed the final ordering of 8 of 10 cases relative to physical-only ranking. The best-ranked physical case was the oral tongue/floor-of-mouth horseshoe case, but it dropped from rank 1 physically to rank 3 with the anatomical sink included. In contrast, the right sinonasal/maxillary bulky ellipsoid improved from physical rank 2 to biological rank 1. Five cases acquired biology-only OAR reinterpretations, all involving brainstem D_2 . The full 10-case rank-change table is shown in Table 2.

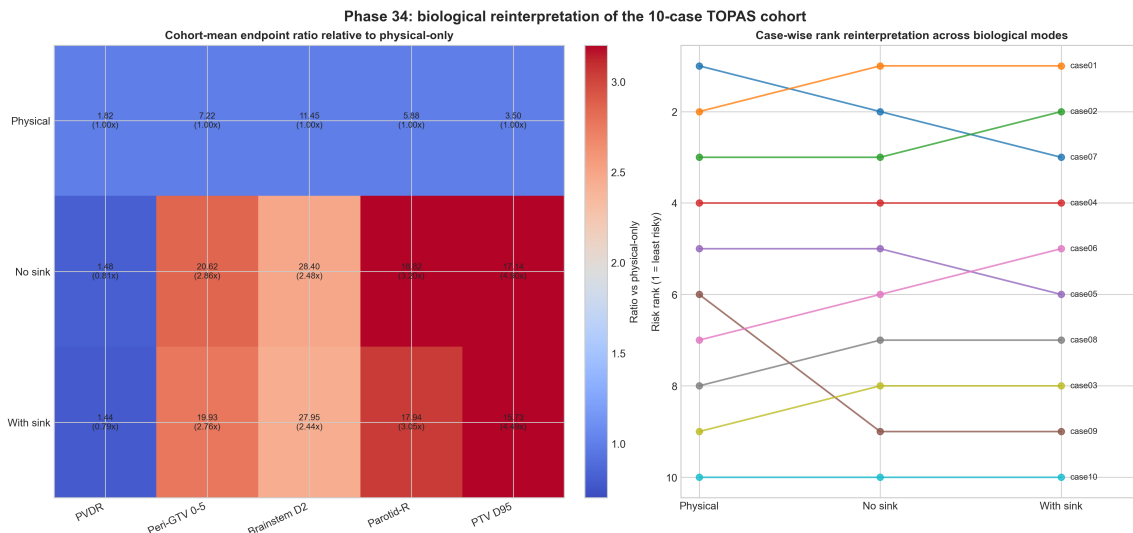


Figure 2: **Biological reinterpretation of the site-specific cohort.** The same physical cohort was rescored under biology-informed post-processing with and without anatomical vascular uptake. Plan ordering changed in most cases, and spill and OAR burden rose substantially relative to the physical-only view.

Table 2: **Rank changes across the 10-case cohort after biology-informed reinterpretation.** Negative shift values indicate improvement relative to the physical-only rank.

Case	Site / tumour model	Physical rank	No-sink rank	Anatomical sink rank	Sink vs physical shift	vs Biology-only reinterpretation
07	Oral tongue / floor-of-mouth horseshoe	1	2	3	+2	None
01	Right sinonasal / maxillary bulky ellipsoid	2	1	1	-1	None
02	Left maxillary crescent abutting orbit	3	3	2	-1	None
04	Laryngo-hypopharyngeal elongated cylinder	4	4	4	0	Brainstem D_2
05	Parapharyngeal / prestyloid deep-space mass	5	5	6	+1	Brainstem D_2
09	Deep parotid / infratemporal bulky mass	6	9	9	+3	Brainstem D_2
06	Buccal mucosa / cheek infiltrative crescent	7	6	5	-2	None
08	Bulky nodal conglomerate with central necrosis	8	7	7	-1	None
03	Base-of-tongue / oropharynx bilobed mass	9	8	8	-1	Brainstem D_2
10	Composite oropharynx + upper-neck bulky tumour	10	10	10	0	Brainstem D_2

1.3 Uncertainty analysis

Repeat-run uncertainty was evaluated on a six-case subset using three seeds and three history scales, giving nine repeated transport combinations. Endpoint-level sink effects remained visible after uncertainty was introduced. Across the evaluated case-endpoint set, 40 of 42 sink-driven endpoint deltas exceeded the combined Monte Carlo plus uptake-sensitivity 95% noise band. The strongest repeated signals were observed for PVDR, peri-GTV spill, and right parotid mean. However, no case showed a noise-qualified sink-driven rank change in this subset, indicating that the sink effect was more stable as an endpoint-level modifier than as a rank-reversal mechanism (Figure 3).

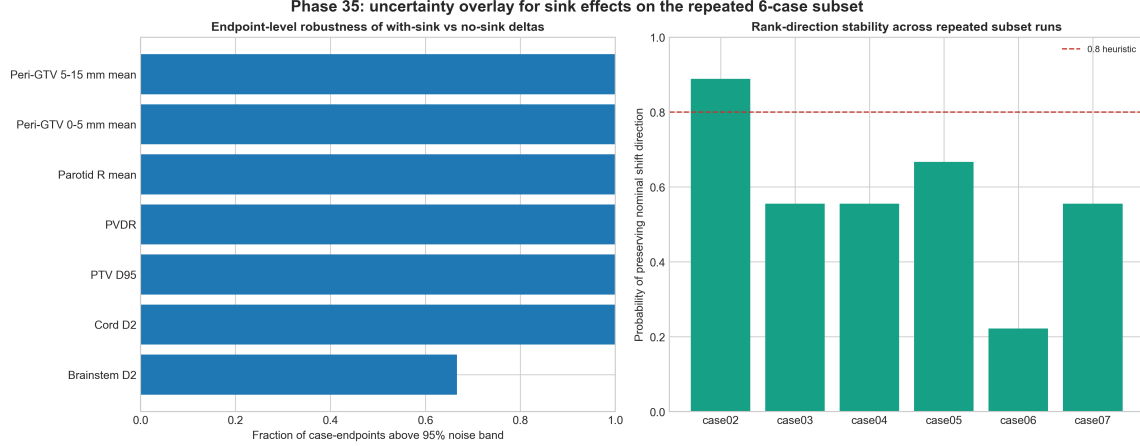


Figure 3: **Uncertainty robustness.** Repeated transport and biology solves preserved most sink-driven endpoint shifts beyond the combined 95% noise band, but did not produce stable sink-driven rank reversals in the repeated six-case subset.

1.4 Vascular-sink falsification

The anatomical sink remained distinguishable from stronger surrogate controls. In the repeated subset, 222 of 288 true-versus-surrogate endpoint differences exceeded the combined 95% noise band. The strongest repeated separations were observed for the no-sink comparator (42/48 endpoints), uniform body washout (42/48), and local vessel dropout near the tumour (41/48). Superior-inferior and anterior-posterior flips remained weaker but still measurable, with 32/48 and 31/48 endpoint differences above the same threshold, respectively. Only 2 of 36 true-versus-surrogate rank comparisons were noise-qualified, and both occurred against the mass-matched uniform washout control. Thus, the anatomical sink was reproducibly different from generic and spatially perturbed alternatives at the endpoint level, but it remained a weak rank-flipping signal (Table 3; Figure 4).

Table 3: **Summary of stronger falsification results in the repeated subset.**

Comparator	Endpoints above 95% noise band	Noise-qualified rank changes
No sink	42/48	0/6
Uniform body washout (mass matched)	42/48	2/6
Local vessel dropout within 20 mm of tumour	41/48	0/6
Randomized anatomy-constrained displacement	34/48	0/6
Superior-inferior flip	32/48	0/6
Anterior-posterior flip	31/48	0/6

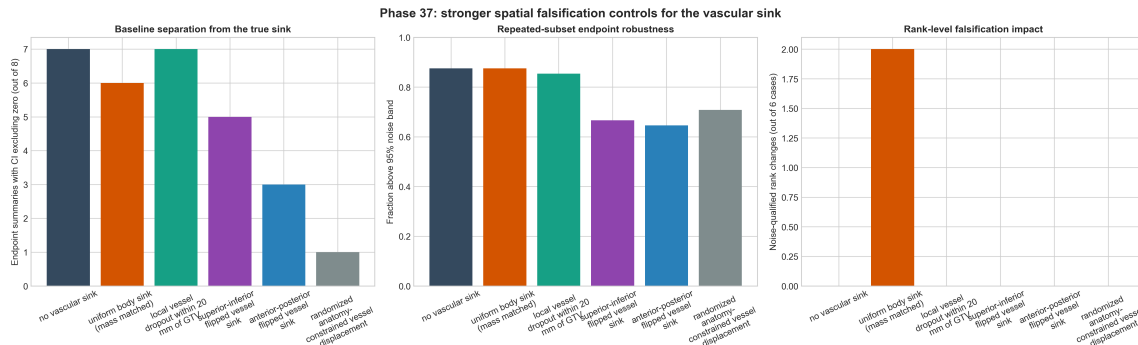


Figure 4: **Stronger spatial falsification controls.** The anatomical sink remained distinguishable from multiple perturbed or redistributed washout fields, especially from local vessel dropout and uniform body washout, even after repeat-run uncertainty was included.

1.5 Summary

Taken together, the cohort results support three points. First, the physical TOPAS-derived photon lattice plans were reproducibly generated across 10 anatomically distinct synthetic H&N cases. Second, biology-informed post-processing materially changed case interpretation, including rank order and OAR burden. Third, the vascular-sink term behaved as a measurable endpoint-level modifier that remained distinguishable from several stronger surrogate controls, although it did not produce broad noise-qualified rank reversals.